ONE HUNDRED BOOKS IN TEN ACTS

A Genealogy of Technological Optimism

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THE SERIES

This is an intellectual genealogy of technological optimism in 100 books, ordered as an argument in ten acts. It begins with the theology that the idea of progress grew out of, and the psychology of the people who act on it. Then it follows how knowledge grows, how markets and institutions turn it into growth, how innovation is built and financed, and the attempt to engineer away every remaining constraint.

About three quarters of the books make the case. The other quarter are the strongest arguments against it. Each critic sits next to the claims it attacks, and critiques of technology itself are held for the final act. The last book, David Deutsch's *The Beginning of Infinity*, has to try to answer the critics.

Roughly 40 titles are canonical and 60 are the sources the canon was built from. The order is the part most worth arguing with. Years are first publication in the original language.

ACTS

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Listen at jrslotkin.github.io/genealogy-of-optimism. Tap any title to find it there.
Set in Newsreader and Geist.

Why should history go anywhere?

Progress as cosmic destiny, then as theology turned into a research program, then the dispute over whether it is religion in disguise.

1 ## The Phenomenon of Man

Pierre Teilhard de Chardin 1955

A Jesuit paleontologist reads evolution as one movement toward greater complexity and consciousness, ending in a planetary mind.

2 ## The Philosophy of the Common Task

Nikolai Fedorov 1906

A Moscow librarian argues that humanity's shared moral duty is to defeat death and resurrect every ancestor through science. Tsiolkovsky studied in his reading room, and Russian cosmism starts here.

3 ## New Atlantis

Francis Bacon 1626

Salomon's House is a detailed blueprint for a state-funded research institution, written before one existed. A template for the Royal Society and, at some distance, for every lab since.

4 ## Sketch for a Historical Picture of the Progress of the Human Mind

Nicolas de Condorcet 1795

Written in hiding during the Terror, shortly before he died in custody. Argues that improvement in knowledge, equality, and even lifespan has no fixed limit.

5 ## History of the Idea of Progress

Robert Nisbet 1980

Traces progress from the Greeks through Augustine to the moderns and argues it has always rested on something close to faith. Sets up the dispute in the next two books.

6 ## Meaning in History

Karl Löwith 1949

The charge: modern faith in progress is Christian eschatology with God removed. It applies to the way AGI is now described.

7 **The Legitimacy of the Modern Age**

Hans Blumenberg 1966

The reply to Löwith. Curiosity and human self-assertion are modernity's own achievements and owe nothing to the theology they replaced.

8 **The Protestant Ethic and the Spirit of Capitalism**

Max Weber 1905

Capitalist discipline grew out of religious anxiety about salvation. It ends with the iron cage, a system that keeps running after the belief that built it has gone.

9 **The Spirit of Democratic Capitalism**

Michael Novak 1982

A Catholic theologian's case that democratic capitalism is a moral and spiritual achievement. The theological defense its critics seldom engage.

10 **Frankenstein; or, The Modern Prometheus**

Mary Shelley 1818

A maker abandons what he made, and the creation asks what it is owed.

What kind of creature is building the future?

Ambition, mortality, imitation, error, and how badly experts predict.

11 **Thus Spoke Zarathustra**

Friedrich Nietzsche 1883

Man as something to be overcome. Transhumanism and accelerationism both draw on it, usually without saying so.

12 **The Denial of Death**

Ernest Becker 1973

Civilization as an immortality project built to manage the fear of death. Explains a great deal about why people build, and raises a hard question about longevity research.

13 **Things Hidden Since the Foundation of the World**

René Girard, with Jean-Michel Oughourlian and Guy Lefort 1978

Mimetic desire and the scapegoat mechanism, set out as dialogues. Thiel encountered Girard at Stanford, and Zero to One's view of competition draws on it.

14 **Notes from Underground**

Fyodor Dostoevsky 1864

A narrator who would rather act against his own interest than live as a piano key in a rationally planned world. The first serious argument that people will reject a utopia designed for their benefit.

15 **Thinking, Fast and Slow**

Daniel Kahneman 2011

The catalog of systematic error in human judgment. Read the priming chapters with the replication record in mind; the core heuristics-and-biases findings have held up better.

16 **Rationality for Mortals**

Gerd Gigerenzer 2008

Under real uncertainty, simple heuristics often outperform optimization. Closer to how founders and investors actually decide, and a direct challenge to Kahneman's framing.

17 ## The Secret of Our Success

Joseph Henrich 2015

Humans succeed through accumulated culture, and innovation scales with the size and connectedness of the population sharing it. The anthropological base for why networks and cities matter later.

18 ## Origins of Genius

Dean Keith Simonton 1999

Creativity treated as Darwinian variation and selection. The best predictor of a creator's hits is the number of attempts, the same distribution venture portfolios live in.

19 ## Expert Political Judgment

Philip E. Tetlock 2005

Two decades of expert forecasts scored against outcomes. Foxes beat hedgehogs, and simple statistical extrapolation beat most of the experts.

20 ## Foundation

Isaac Asimov 1951

Placed right after Tetlock: a mathematician who can forecast the fall of a galactic civilization. Psychohistory has had a long afterlife among people who build companies.

How can a creature like that know anything?

Error correction, paradigms, tacit and dispersed knowledge, emergence, feedback, and design.

21 **Science and the Modern World**

Alfred North Whitehead 1925

Names the method of invention as the nineteenth century's greatest invention. Also introduces the fallacy of misplaced concreteness, a warning against mistaking models for the world.

22 **Conjectures and Refutations**

Karl Popper 1963

Knowledge grows through bold guesses and criticism, never through confirmation. The epistemology underneath Deutsch, who closes the list.

23 **The Structure of Scientific Revolutions**

Thomas S. Kuhn 1962

Science moves through paradigms that decide which questions count and which anomalies get ignored. The complication Popper never fully answered.

24 **The Tacit Dimension**

Michael Polanyi 1966

We can know more than we can tell. Explains why expertise resisted codification for decades, and why systems that learn from examples finally got around it.

25 **Individualism and Economic Order**

Friedrich A. Hayek 1948

Contains "The Use of Knowledge in Society" (1945): prices coordinate knowledge that no single mind holds.

26 **Micromotives and Macrobehavior**

Thomas C. Schelling 1978

Mild individual preferences produce aggregate outcomes nobody chose, from segregated neighborhoods to tipping points. Adoption dynamics and network effects, worked out before the startup vocabulary existed.

27 **The Human Use of Human Beings**

Norbert Wiener 1950

Feedback and control as the logic of machines and societies, from the founder of cybernetics, with an early warning about automation and labor.

28 **The Sciences of the Artificial**

Herbert A. Simon 1969

Design, engineering, and organizations treated as one science of the artificial under bounded rationality. Simon co-wrote one of the first AI programs, and this is his clearest statement of intelligence as search and design.

Can markets turn ignorance into progress?

Uncertainty, the entrepreneur, the firm, the moral case and its strongest critic, and the institutions that decide whether any of it compounds.

29 **An Inquiry into the Nature and Causes of the Wealth of Nations**

Adam Smith 1776

Markets as a way of coordinating strangers. Read it for the claim that the division of labor is limited by the extent of the market, an argument about scale.

30 **Risk, Uncertainty and Profit**

Frank H. Knight 1921

Separates measurable risk from true uncertainty and argues that profit exists only because of the second. Venture investing is built on that distinction.

31 **The Theory of Economic Development**

Joseph A. Schumpeter 1911

The entrepreneur first appears as the agent of “new combinations” who breaks equilibrium. Capitalism, Socialism and Democracy is the famous book; this is the source.

32 **The Firm, the Market, and the Law**

Ronald H. Coase 1988

Collects “The Nature of the Firm” (1937): firms exist because using markets has costs. Each drop in coordination cost, including the one AI is producing, moves the boundary of the firm.

33 **Bourgeois Dignity**

Deirdre N. McCloskey 2010

The Great Enrichment followed a change in how societies talked about and honored commerce and innovation, more than any shift in capital, trade, or institutions. The middle volume of her trilogy.

34 **Atlas Shrugged**

Ayn Rand 1957

The producer as moral hero. The novel builder culture returns to most. Read it against Notes from Underground, twenty books earlier.

35 **Anarchy, State, and Utopia**

Robert Nozick 1974

The rigorous case for a minimal state, ending in a “framework for utopia” where people sort into communities of their choosing. Network-state and charter-city arguments are versions of Part III.

36 **The Great Transformation**

Karl Polanyi 1944

Self-regulating markets were built by states, and society pushes back when they outrun it. Michael Polanyi’s brother, placed directly after Rand and Nozick.

37 **The Economy of Cities**

Jane Jacobs 1969

New work grows by adding to old work, in dense places. Also argues, against the textbook sequence, that cities came before agriculture.

38 **Institutions, Institutional Change and Economic Performance**

Douglass C. North 1990

Institutions are the rules of the game and set the incentives for everything else. The source Why Nations Fail builds on.

39 **The Rise and Decline of Nations**

Mancur Olson 1982

Stable societies accumulate interest groups that block change, which helps explain why defeated or disrupted nations often grow fastest afterward.

40 **Exit, Voice, and Loyalty**

Albert O. Hirschman 1970

Dissatisfied members can leave or speak up, and each option weakens the other. The analytic root of every exit argument later in the list.

Why did growth happen, and why did it slow?

The Great Enrichment, the Malthus debate, and the stagnation debate.

41 ## A Culture of Growth

Joel Mokyr 2016

A competitive market for ideas among Europe's scholars, the Republic of Letters, set up the Industrial Revolution. Mokyr shared the 2025 Nobel for this line of work.

42 ## Knowledge and the Wealth of Nations

David Warsh 2006

The story of Paul Romer's point that ideas are nonrival, so growth can keep compounding.

43 ## An Essay on the Principle of Population

Thomas Robert Malthus 1798

Written against Condorcet and Godwin: population outruns food, so improvement is temporary. Wrong about the outcome for two centuries and still the template for every scarcity argument.

44 ## The Ultimate Resource 2

Julian L. Simon 1996

A revision of The Ultimate Resource (1981). Human ingenuity is the resource that does not run out, and Simon's 1980 bet with Paul Ehrlich on commodity prices gives the Malthus rebuttal a scorecard.

45 ## The Rational Optimist

Matt Ridley 2010

Exchange and specialization as the engine, with ideas recombining the way genes do.

46 ## Stubborn Attachments

Tyler Cowen 2018

Sustainable growth deserves near-absolute moral priority because its benefits compound across generations. The moral claim the rest of the optimist case usually leaves implicit.

47 **The Cost Disease**

William J. Baumol 2012

Labor-intensive services get relatively more expensive as productivity rises elsewhere. Health care and education are the standard cases, and whether AI breaks the pattern is an open question.

48 **The Rise and Fall of American Growth**

Robert J. Gordon 2016

The inventions of 1870 to 1970 (electricity, the internal combustion engine, indoor plumbing, antibiotics) changed daily life more than anything since. The strongest case that the big gains are behind us.

49 **Where Is My Flying Car?**

J. Storrs Hall 2018

The same question with the opposite answer: per-capita energy use flattened in the 1970s and regulation did the rest.

50 **Abundance**

Ezra Klein and Derek Thompson 2025

The case for building, made from the center left: permitting, process, and a governing culture organized around veto points. Significant partly because of which coalition it comes from.

51 **Breakneck** *China's Quest to Engineer the Future*

Dan Wang 2025

China as an engineering state that builds and the United States as a lawyerly society that litigates, with the costs of each. Grounds the stagnation debate in a live comparison.

What is innovation made of?

Routines, combination, diffusion, and the relationship between science and use.

52 **An Evolutionary Theory of Economic Change**

Richard R. Nelson and Sidney G. Winter 1982

Firms as bundles of routines that vary and get selected, in place of the optimizing firm of textbook economics. The foundation of evolutionary economics.

53 **Inside the Black Box** *Technology and Economics*

Nathan Rosenberg 1982

Opens the box economists drew around technology. Science often follows engineering, and most of the value arrives through incremental improvement after the invention.

54 **The Nature of Technology** *What It Is and How It Evolves*

W. Brian Arthur 2009

Technologies are assembled from other technologies and evolve by combination. An account of why innovation speeds up as the stock of components grows.

55 **Diffusion of Innovations**

Everett M. Rogers 1962

The S-curve and adopter categories that Crossing the Chasm and every adoption chart since are built on, drawn partly from studies of Iowa farmers adopting hybrid corn.

56 **Science, the Endless Frontier**

Vannevar Bush 1945

The report to the president that shaped the postwar American research system and its linear model running from basic science to products.

57 **Pasteur's Quadrant** *Basic Science and Technological Innovation*

Donald E. Stokes 1997

Takes Bush's linear model apart. Pasteur pursued fundamental understanding because he wanted to solve practical problems, and much of the most productive research sits in that quadrant.

Technology meets the real world

Large systems and the institutions that built them, then who paid, who benefited, what local knowledge was lost, and how such systems fail. The act moves from triumph to failure.

58 **Networks of Power** *Electrification in Western Society, 1880–1930*

Thomas P. Hughes 1983

Electrification compared across Chicago, Berlin, and London, with Edison and Insull as system builders. The closest historical analog to the AI infrastructure buildout.

59 **The Making of the Atomic Bomb**

Richard Rhodes 1986

Physics converted into a weapon at wartime speed, with the science, the institutions, and the moral weight in one book. Widely read inside the frontier AI labs.

60 **The Idea Factory** *Bell Labs and the Great Age of American Innovation*

Jon Gertner 2012

The transistor, information theory, the laser, and the management model behind them. Also a study of what a regulated monopoly could afford to fund.

61 **The Dream Machine**

M. Mitchell Waldrop 2001

Licklider's idea of human-computer symbiosis and how ARPA turned it into interactive computing and the network that became the internet. A case study in how one funder with a clear vision can redirect a field.

62 **Regional Advantage** *Culture and Competition in Silicon Valley and Route 128*

AnnaLee Saxenian 1994

Why the Valley's open, networked culture outcompeted Route 128's vertically integrated firms.

63 **The Box**

Marc Levinson 2006

A steel box rewired global trade by cutting the cost of the whole system, in ways the early players did not foresee. The model for recognizing technologies whose impact is mostly second-order.

64 **Chip War** *The Fight for the World's Most Critical Technology*

Chris Miller 2022

How semiconductors became the strategic chokepoint, from Fairchild to TSMC to export controls.

65 **The Entrepreneurial State**

Mariana Mazzucato 2013

Public money funded the riskiest underlying science (the internet, GPS, touchscreens) before private capital commercialized it.

66 **Power and Progress**

Daron Acemoglu and Simon Johnson 2023

Gains from technology are shared only when workers and institutions can claim them, and the direction of innovation is a choice. Both authors shared the 2024 Nobel.

67 **Seeing Like a State**

James C. Scott 1998

High-modernist schemes fail because they erase the local, practical knowledge they cannot see. Hayek's argument aimed at the state, and just as applicable to platforms.

68 **Normal Accidents** *Living with High-Risk Technologies*

Charles Perrow 1984

In tightly coupled, interactively complex systems, some catastrophic failures are built into the design. Written after Three Mile Island and directly relevant to autonomous systems and grids.

69 **The Whale and the Reactor**

Langdon Winner 1986

Contains “Do Artifacts Have Politics?”: design choices settle questions of power before any law is written.

70 **The Collapse of Complex Societies**

Joseph A. Tainter 1988

Societies solve problems by adding complexity until the returns to complexity turn negative. A theory of collapse with no villain.

Who pays for the future?

Finance under true uncertainty: bubbles, power laws, bet sizing, value capture, and fraud. Venture arrives late on purpose, after the ideas it rests on.

71 **VC** *An American History*

Tom Nicholas 2019

The rigorous history, starting with nineteenth-century New Bedford whaling, whose return distribution looked much like venture's.

72 **Doing Capitalism in the Innovation Economy**

William H. Janeway 2012

Innovation as a three-player game among the state, speculative finance, and the market economy, by an economist who spent decades investing in technology at Warburg Pincus.

73 **Technological Revolutions and Financial Capital** *The Dynamics of Bubbles and Golden Ages*

Carlota Perez 2002

Each technological revolution runs through an installation phase financed by a bubble, a crash, and then a deployment period. The framework most people reach for when reading the current AI capital cycle.

74 **Engines That Move Markets**

Alasdair Nairn 2002

The same revolutions from the investor's side, from railways to the internet. Most investors in transformative technologies lost money even when the technology won.

75 **Fortune's Formula**

William Poundstone 2005

The Kelly criterion, from John Kelly at Bell Labs and Claude Shannon and Ed Thorp at MIT to casinos and Wall Street. Bet sizing when the edge is real and the variance is enormous.

76 **The Innovator's Dilemma**

Clayton M. Christensen 1997

Well-managed incumbents lose because they serve their best customers well. Widely cited and often misapplied; the original is more careful than its summaries.

77 **7 Powers** *The Foundations of Business Strategy*

Hamilton Helmer 2016

Seven sources of durable advantage, and a clean separation between a technology mattering and a company capturing its value.

78 **The Power Law** *Venture Capital and the Making of the New Future*

Sebastian Mallaby 2022

Venture as a distinct form of finance built around outliers, from Arthur Rock and Fairchild forward.

79 **High Output Management**

Andrew S. Grove 1983

A manager's output is the output of the organization, and Intel's methods for raising it.

80 **Zero to One**

Peter Thiel with Blake Masters 2014

Definite optimism, secrets, and monopoly. Read here, after Girard, Knight, and Schumpeter, it reads as a compression of the tradition behind it.

81 **Bad Blood**

John Carreyrou 2018

Directly after Thiel: a company whose secret was fabricated, and the narrative, board prestige, and missing diligence that kept it going.

The attempt to escape every constraint

Biology, gravity, matter, the state, and finally human intelligence, each treated as a limit to engineer away, with the sharpest local objections placed in line.

82 **Daedalus, or Science and the Future**

J.B.S. Haldane 1923

A Cambridge lecture predicting ectogenesis and the deliberate engineering of human biology. An opening document of transhumanism and one of Huxley's sources for Brave New World.

83 **The World, the Flesh and the Devil**

J.D. Bernal 1929

A young crystallographer imagines space habitats, now called Bernal spheres, and the redesign of the human body and mind. Much of today's frontier agenda is already in it.

84 **The Abolition of Man**

C.S. Lewis 1943

Each new power over nature turns out to be power some people hold over others, including over future generations. Haldane later attacked Lewis's view of science in print, and Lewis drafted a reply.

85 **The High Frontier** *Human Colonies in Space*

Gerard K. O'Neill 1976

A Princeton physicist's engineering case for large rotating colonies built from lunar and asteroid material. Blue Origin's stated long-term vision is built on it.

86 **The Human Condition**

Hannah Arendt 1958

Opens with Sputnik and the wish to escape the earthbound human condition, then asks what happens to action and meaning in a society of laborers without labor.

87 **Engines of Creation** *The Coming Era of Nanotechnology*

K. Eric Drexler 1986

Molecular manufacturing and the assembler. Much of the nanotech program has not arrived; its reasoning about exponential capability and its dangers carried straight into AI.

88 **The Sovereign Individual**

James Dale Davidson and William Rees-Mogg 1997

Predicted that digital commerce and encryption would loosen the nation-state's hold on mobile capital and talent. Parts read as prophecy and parts as wishful thinking.

89 **Anti-Oedipus** *Capitalism and Schizophrenia*

Gilles Deleuze and Félix Guattari 1972

Capitalism as a machine that dissolves every fixed code, with a passage suggesting the way out is to go further into the process. Later accelerationists treat that passage as their starting point.

90 **Fanged Noumena** *Collected Writings 1987–2007*

Nick Land 2011

The Warwick and CCRU writings that recast acceleration as capital and technology assembling an intelligence of their own. e/acc descends from these essays, which are stranger and darker than e/acc.

91 **Erewhon**

Samuel Butler 1872

After all that futurism, a Victorian satire whose “Book of the Machines” argues that machines are evolving and should be destroyed before they surpass us.

92 **Summa Technologiae**

Stanisław Lem 1964

A novelist’s nonfiction forecast of virtual reality, artificial intelligence, and technological evolution. Not translated into English until 2013.

93 **Mind Children** *The Future of Robot and Human Intelligence*

Hans Moravec 1988

Extrapolates computing power against estimates of brain capacity to forecast human-level machines, and contains what became known as Moravec’s paradox. The foundation Kurzweil built on.

94 **The Singularity Is Nearer**

Ray Kurzweil 2024

The exponential case restated nineteen years after The Singularity Is Near, with its central timelines unchanged.

95 **The Scaling Era** *An Oral History of AI, 2019–2025*

Dwarkesh Patel with Gavin Leech 2025

The people building frontier models, in their own words, organized as a history of the scaling hypothesis.

The trial

Critiques of technology as such, held to the end so they meet the whole argument, then the answer.

96 **The Technological Society**

Jacques Ellul 1954

Technique, the search for the most efficient method in every domain, becomes autonomous and sets its own ends.

97 **The Question Concerning Technology**

Martin Heidegger 1954

Modern technology reveals the world, people included, as standing-reserve: resources waiting to be ordered and optimized. Short, difficult, and the root of most serious critiques since.

98 **The Imperative of Responsibility**

Hans Jonas 1979

Power on this scale demands an ethics of responsibility toward people not yet born, and a rule that gives the prophecy of doom more weight than the prophecy of bliss. The philosophical root of the precautionary principle.

99 **If Anyone Builds It, Everyone Dies**

Eliezer Yudkowsky and Nate Soares 2025

Building superhuman AI with anything like current methods ends in human extinction, argued without hedging.

100 **The Beginning of Infinity**

David Deutsch 2011

Problems are inevitable, problems are soluble, and knowledge can grow without bound. Placed last so it has to answer everything before it.